Additional Exercise 4 · Exam type

T3 Additional Exercise 4 – CEMED, statements 2016 and supplier discount

Original statement · T3_Additional_EXERCISES.pdf, p. 5

CERÁMICAS DEL MEDITERRÁNEO (CEMED) manufactures and distributes ceramic tiles for construction in Valencia.

- CEMED was created on 1/01/2016 by three partners, each contributing 7 m.u., fully in cash.
- On 1/02/2016 it bought a piece of land worth 10 m.u., paying 50% in cash and leaving 50% to be paid one year later. On the same date it bought the factory building, paying 6 m.u. in cash and financing this amount with a mortgage at 5% annual interest for ten years. The factory is amortised over 20 years, linearly, with zero residual value. Each 31 December the company returns 10% of the mortgage principal plus the interest accrued in the year.
- On 1/04/2016 it obtained an additional bank loan of 5 m.u. at 6% annual interest, returning 20% of the principal each semester for the next 2½ years (first repayment on 1/10/2016, second on 1/04/2017, and so on). Interest is paid semiannually, calculated on the amount actually due in that semester, the day after the end of the semester.
- With this loan, on the same date, it bought machinery for 3 m.u. with 60 days payment delay; useful life 10 years, straight-line, zero residual value.
- On 1/07/2016 it acquired 150 tons of raw material at 0.03 m.u./ton. Half of it was used in producing the products sold. 60% of the total cost is paid to the supplier at 90 days; the other 40% remains unpaid at the end of the year.
- Sales revenue: 50 m.u., of which only 70% had been collected at 31/12/2016.
- Personnel plus Social Security: 0.8 m.u., paid in full in 2016.
- Corporation tax: 30%, settled by calendar year and paid on 15/01/2017.
- The company keeps 40% of annual net profits as reserves; the dividends are paid on 31/12/2016.

Required: (1) the income statement and the balance sheet at 24:00 on 31/12/2016, explaining the intermediate calculations; (2) the changes in the income statement and the balance sheet if the company had negotiated a 5% discount on the total price of the raw materials in exchange for paying in cash the 60% agreed at 90 days. In which case is the profit higher, and why?

Phase 1 — Setup: data, timeline and hypotheses

Date	Event	Cash
1/01	Capital $3 \times 7 = 21$	+21
1/02	Land 10: 50% paid, 50% due 1/02/2017	−5
1/02	Building 6, financed by a mortgage of 6	+6 −6
1/04	Bank loan 5 at 6%	+5
1/04	Machinery 3 bought, paid 60 days later (31/05)	−3 (on 31/05)
1/07	Raw material $150 \times 0.03 = 4.5$	—
29/09	60% of the raw material paid (90 days)	−2.7
1/10	Loan: first instalment (20% of 5) and interest of the first semester	−1 −0.15
31/12	Mortgage: 10% of principal and interest of the year	−0.6 −0.275
31/12	Dividends (60% of net income)	−19.3053
15/01/2017	Tax paid (after the closing)	—

Hypotheses:

1. The building is used from 1/02 (11 months) and the machinery from 1/04 (9 months). Land is not depreciated.
2. The mortgage interest of 2016 runs from 1/02 to 31/12 (11 months) on the 6 m.u. outstanding.
3. Loan semesters: 1/04-30/09 (principal 5) and 1/10-31/03 (principal 4). The interest of the second semester is paid on 1/04/2017, so the part from 1/10 to 31/12 is an **accrued, unpaid** expense of 2016.
4. Results to four decimals; exact fractions were used and the last digit is rounded.

Phase 2 — Frames: classification of every operation

Operation	Income statement 2016	Cash 2016	Open at 31/12/2016
Capital	—	+21	Capital 21
Land	—	−5	Land 10; land payable 5 (current)
Building / mortgage	Depreciation 0.275; interest 0.275	+6 −6 −0.6 −0.275	Building $6 - 0.275$; mortgage 5.4 (0.6 current, 4.8 long-term)
Bank loan	Interest $0.15 + 0.06$	+5 −1 −0.15	Loan 4 (2 current, 2 long-term); interest payable 0.06
Machinery	Depreciation 0.225	−3	Machinery $3 - 0.225$
Raw material	COGS 2.25	−2.7	Inventory 2.25; suppliers 1.8
Sales	Revenue 50	+35	Receivables 15

Operation	Income statement 2016	Cash 2016	Open at 31/12/2016
Personnel + SS	0.8	-0.8	–
Tax	13.7895	0	Taxes payable 13.7895
Dividends	– (distribution)	-19.3053	Reserves 12.8702

Phase 3 – Derivation (requirement 1)

Step 1. Depreciation

$$D_{\text{building}} = \frac{6-0}{20} \times \frac{11}{12} = 0.3 \times 0.9167 = 0.2750 \text{ m.u.}$$

$$D_{\text{machinery}} = \frac{3-0}{10} \times \frac{9}{12} = 0.3 \times 0.75 = 0.2250 \text{ m.u.}$$

Total depreciation: $0.275 + 0.225 = 0.5$ m.u.

Step 2. Mortgage

Interest of 2016 (1/02 to 31/12, principal 6):

$$I_{\text{mortgage}} = 6 \times 0.05 \times \frac{11}{12} = 0.2750 \text{ m.u.} \quad (\text{paid on } 31/12)$$

Principal returned on 31/12: $0.10 \times 6 = 0.6$; outstanding $6 - 0.6 = 5.4$. The next 0.6, due 31/12/2017, is current; the remaining $5.4 - 0.6 = 4.8$ is long-term.

Step 3. Bank loan

Each instalment is $0.20 \times 5 = 1$ m.u. (five instalments: 1/10/2016, 1/04/2017, 1/10/2017, 1/04/2018, 1/10/2018).

- First semester (1/04-30/09, principal 5): $I_1 = 5 \times 0.06 \times \frac{6}{12} = 0.15$, paid on 1/10/2016 together with the first instalment.
- Second semester (1/10/2016-31/03/2017, principal 4): the 2016 part is October-December, $I_2 = 4 \times 0.06 \times \frac{3}{12} = 0.06$, **accrued but unpaid** (it will be paid on 1/04/2017).

Interest of the loan in 2016: $0.15 + 0.06 = 0.21$. Principal outstanding at 31/12: $5 - 1 = 4$, of which the instalments of 1/04/2017 and 1/10/2017 (2) are current and those of 2018 (2) are long-term.

Step 4. Raw material, cost of goods sold and suppliers

$$\text{Purchase} = 150 \times 0.03 = 4.5 \text{ m.u.}, \qquad \text{COGS} = \frac{4.5}{2} = 2.25, \qquad \text{Inventory} = 2.25$$

Paid at 90 days: $0.60 \times 4.5 = 2.7$; unpaid: $0.40 \times 4.5 = 1.8$ (suppliers, current).

Step 5. Sales

Collected $0.70 \times 50 = 35$; receivables $50 - 35 = 15$.

Step 6. Income statement 2016

Line	Computation	m.u.
Sales revenue		50.0000
– Cost of goods sold	Step 4	2.2500
= Gross profit		**47.7500**
– Personnel and Social Security		0.8000
– Depreciation	Step 1	0.5000
= Income from operations	$47.75 - 1.3$	**46.4500**
– Interest (mortgage 0.275; loan 0.21)	Steps 2-3	0.4850
= Income before taxes	$46.45 - 0.485$	**45.9650**
– Income taxes (30%)	0.30×45.965	13.7895
= Net income	$45.965 - 13.7895$	**32.1755**

Distribution: reserves $= 0.40 \times 32.1755 = 12.8702$; dividends $= 0.60 \times 32.1755 = 19.3053$, paid on 31/12.

Step 7. Cash at 24:00 on 31/12/2016

Inflows	m.u.	Outflows	m.u.
Capital	21.0000	Land (50%)	5.0000
Mortgage	6.0000	Building	6.0000
Bank loan	5.0000	Machinery	3.0000
Sales collected	35.0000	Raw material (60%)	2.7000
		Personnel	0.8000
		Mortgage principal + interest	0.8750
		Loan instalment + interest	1.1500
		Dividends	19.3053
Total	**67.0000**	**Total**	**38.8303**

$$\text{Cash} = 67 - 38.8303 = 28.1697 \text{ m.u.}$$

Step 8. Balance sheet at 24:00 on 31/12/2016

Assets	m.u.	Liabilities and equity	m.u.
Land	10.0000	Capital	21.0000
Building (6 − 0.275)	5.7250	Reserves (40% of net income)	12.8702
Machinery (3 − 0.225)	2.7750	**Equity**	**33.8702**
Non-current assets	**18.5000**	Mortgage, long-term part	4.8000
Inventory (raw material)	2.2500	Bank loan, long-term part	2.0000
Accounts receivable	15.0000	**Long-term liabilities**	**6.8000**
Cash	28.1697	Land payable (1/02/2017)	5.0000
Current assets	**45.4197**	Mortgage, current part	0.6000
		Bank loan, current part	2.0000
		Interest payable	0.0600
		Suppliers	1.8000
		Taxes payable (15/01/2017)	13.7895
		Current liabilities	**23.2495**
Total assets	**63.9197**	**Total liabilities and equity**	**63.9197**

Phase 3b – Derivation (requirement 2: 5% discount, 60% paid in cash)

Step 9. New purchase price and payments

$$\text{Price} = 4.5 \times (1 - 0.05) = 4.275 \text{ m.u.}$$

- Paid in cash on 1/07: $0.60 \times 4.275 = 2.565$ (instead of 2.7 at 90 days).
- Unpaid at 31/12: $0.40 \times 4.275 = 1.71$ (instead of 1.8).
- Consumed (half): $\text{COGS} = 4.275/2 = 2.1375$; inventory 2.1375 (both instead of 2.25).

Step 10. New income statement

Only the cost of goods sold changes before taxes: $\Delta\text{COGS} = 2.1375 - 2.25 = -0.1125$.

Line	Requirement 1	Requirement 2	Change
Cost of goods sold	2.2500	2.1375	−0.1125
Income before taxes	45.9650	46.0775	+0.1125
Income taxes (30%)	13.7895	13.8233	+0.0338
Net income	**32.1755**	**32.2543**	**+0.0788**
Dividends (60%)	19.3053	19.3526	+0.0473
Reserves (40%)	12.8702	12.9017	+0.0315

The net change is the after-tax saving: $0.1125 \times (1 - 0.30) = 0.07875$.

Step 11. New balance sheet items

Item	Change	Reason
Inventory	−0.1125	Half of the material is still in stock, valued at the lower price
Cash	+0.0878	Pays 2.565 instead of 2.7 (+0.135) but pays 0.0473 more dividends
Suppliers	−0.0900	40% of a lower price (1.71 instead of 1.8)
Taxes payable	+0.0338	Higher profit
Reserves	+0.0315	40% of the higher net income
Total assets	**−0.0248**	$-0.1125 + 0.08775$
Total liabilities and equity	**−0.0248**	$-0.09 + 0.03375 + 0.0315$

New totals: $63.9197 - 0.0248 = 63.8950$ on both sides (cash 28.2575).

Phase 4 – Interpretation and checks

- **Balance:** $18.5 + 45.4197 = 63.9197 = 33.8702 + 6.8 + 23.2495$; requirement 2: both sides fall by 0.02475.
- **In which case is the profit higher?** With the **discount**. The discount lowers the cost of the material; the price paid for it is giving up 90 days of **trade credit** [Slide 35], and trade credit has no explicit interest in this problem, so losing it costs nothing in the income statement: it only advances a payment that in both cases falls inside 2016.
- **Why only half of the saving reaches the profit:** only half of the material was consumed. The other half of the saving is in the lower value of the inventory and will reach the income statement when that stock is used.
- **Debt structure** [Slide 36]: liabilities $6.8 + 23.2495 = 30.0495$, equity 33.8702, so $L/E = 0.89$.
- **Exam traps:** (1) the mortgage finances the building: cash +6 and −6 on the same day; (2) the loan interest of October-December is accrued but unpaid; (3) the current part of each loan is what is due before 31/12/2017; (4) the land is not depreciated; (5) taxes are payable on 15/01/2017, so they are a liability, while dividends were paid.

Check against the official key

OFFICIAL SOLUTION, T3_ADDITIONAL_SOLUTIONS.PDF, PP. 19-25

Requirement 1 coincides with the key: cost of goods sold 2.25; personnel 0.8; depreciation 0.5; EBIT 46.45; interest 0.485; income before taxes 45.965; tax 13.79; net income 32.18; dividends 19.31; reserves 12.87; cash 28.17; total 63.92. Requirement 2 also coincides: cost of goods sold 2.14 (2.1375, rounded in the key); net income 32.25; dividends 19.35; cash 28.26; suppliers 1.71; total 63.89. The key's figures show net income rising from 32.18 to 32.25, but the key states no conclusion to the question "in which case are profits higher?"; the reasoning in Phase 4 is ours.

DISCREPANCY WITH THE OFFICIAL SOLUTION (CURRENT PART OF THE DEBT)

The key leaves the whole bank loan (4.0) and the whole mortgage (5.4) as long-term debt, so its current liabilities are 20.65 and its long-term liabilities 9.4. Here the instalments due within twelve months of 31/12/2016 are shown as current: 2.0 of the loan (1/04/2017 and 1/10/2017) and 0.6 of the mortgage (31/12/2017), following the definition of Slide 16. Current liabilities are therefore 23.25 and long-term liabilities 6.8; total liabilities (30.05), equity and total assets are the same.

DIFFERENCES IN THE WORDING OF THE KEY

The key's statement dates the company in 2010, while the official sheet in `sources/` says 2016, which is the year used here. In the key's part 2 the interest formula writes 0.85 where 0.80 is meant (the loan balance of 4 over 5); its value, 0.06, is unchanged.